VAIBHAV AMULANI



Computer Science & Engineering Undergraduate

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LinkedIn | GitHub | Portfolio | LeetCode

PROFESSIONAL SUMMARY

Computer Science undergraduate with a strong foundation in programming, data structures, algorithms, and software engineering. Built and deployed full-stack applications with relational-database backends and practical AI integration. A product-minded problem solver focused on reliable, secure, maintainable software and continuously strengthening engineering fundamentals.

EDUCATION

B.Tech — Computer Science and Engineering **Expected 2028**
Lakshmi Narain College of Technology & Science (LNCTS), Bhopal | Current CGPA: 8.25/10

Class XII — PCM (CBSE) **2024**
Mithi Gobindram Public School | Percentage: 92%

Class X (CBSE) **2022**
Mithi Gobindram Public School | Percentage: 87.7%

TECHNICAL SKILLS

PROGRAMMING LANGUAGES	C++, Java, Python, JavaScript, TypeScript
FRONTEND & MOBILE	React, React Native, Expo, Vite, HTML, CSS, Tailwind CSS, shadcn/ui, TanStack Query
BACKEND & APIs	Node.js, Express.js, REST APIs, API integration, Authentication and authorization, FastAPI
DATABASES & DATA ACCESS	PostgreSQL, SQL, Supabase, MongoDB, Mongoose, Drizzle ORM, Row-Level Security (RLS), Schema Design, Access Control
AI & APPLICATIONS	Google Gemini API, OpenAI API, Streamlit, Rule-Based Diagnostics
DEVELOPER TOOLS & TESTING	Git, GitHub, Vercel, Hostinger, pytest, Vitest, Payment/Webhook Verification, Secure File Uploads, Signed URL Downloads
SYSTEM DESIGN & INFRASTRUCTURE	Docker, System Design Fundamentals, CI/CD, Linux, Containerization, Deployment & Hosting
CORE COMPUTER SCIENCE	Data Structures & Algorithms, OOP, DBMS, Computer Networks, Software Engineering, Machine Learning Fundamentals, Problem Solving

SELECTED PROJECTS

Rays Academy Live Site | GitHub
React, Vite, Tailwind CSS, Supabase, PostgreSQL, Razorpay

- Built and deployed a production website for a real coaching institute, giving it an online presence for its courses, faculty, and institute information.
- Developed a study-material marketplace with a public catalog, free and paid resources, and controlled access to purchased materials.
- Implemented secure Razorpay payment verification with signature validation and idempotent webhook processing for reliable entitlement handling.
- Secured file delivery with Row-Level Security, signed download URLs, and PDF upload validation; covered by 102 automated tests (pgTAP, Vitest).

Nostera — Privacy-Focused Emotional Memory App

COMING SOON ON GOOGLE PLAY

React Native, Expo, TypeScript, Node.js, PostgreSQL, Supabase, AI integration

- Building a mobile archive for preserving meaningful life moments, designed around user-owned memories, privacy-first AI features, and secure data isolation.
- Architecting per-user data isolation with backend and database access controls, plus AI-assisted experiences built around personal memories.

NetSage AI — Network Troubleshooting Assistant

Live Demo | GitHub

Python, Streamlit, Google Gemini API, pytest

- Developed a hybrid network-troubleshooting assistant combining 34 deterministic diagnostic rules with optional AI assistance to help analyze Cisco networking issues.
- Evaluated on a 30-case dataset covering VLANs, OSPF, ACLs, and IPv6; includes JSONL audit logging, human review of outputs, and 31 passing pytest tests.

ToolVerse

Live Demo | GitHub

React, TypeScript, Vite, Tailwind CSS, shadcn/ui, Vercel

- Built and deployed a browser-based collection of 133+ free tools for calculations, text processing, conversions, generators, and developer/student workflows, using client-side processing.

HomeSync — AI-Assisted Household Management

IN DEVELOPMENT GitHub

React 19, Vite, TanStack Query, Tailwind CSS, Express 5, PostgreSQL, Drizzle ORM, Clerk, OpenAI API

- Developing a household SaaS for shared tasks, shopping lists, calendars, meal planning, and maintenance tracking, with Clerk authentication.
- Exploring OpenAI integration for smart shopping lists and meal suggestions from fridge photos (planned, not yet shipped).

CERTIFICATIONS

Programming in Java	NPTEL | 92% | Gold + Elite
Data Structures & Algorithm	NPTEL | 52%
Cisco AICTE Virtual Internship 2026	Cisco Networking Academy | AICTE
Introduction to Modern AI	Cisco Networking Academy
Apply AI: Analyze Customer Reviews	Cisco Networking Academy
Data Analytics Essentials	Cisco Networking Academy
Introduction to IoT and Digital Transformation	Cisco Networking Academy
Vibe Coding - AI Assisted Application Development	NASSCOM | 100% | Gold
Prompt Engineering and ChatGPT	NASSCOM | 85% | Gold

RESEARCH PAPERS

- **AI-based Student Performance Prediction and Early Warning System**
- **Proactive Cyber Threat Prevention Using Ethical Hacking: A Six-Phase Integration Model**

ACHIEVEMENTS

- Secretary, IEEE Student Branch Executive Committee, LNCTS.
- Won Gold Medals in One Act Play and Mime as part of the LNCTS team, which secured Overall First Prize at the 2025 RGPV Nodal Level Art & Drama Competition.

ADDITIONAL INFORMATION

- **Technical Interests:** Backend Engineering, System Design, Database Management, API Development, Cloud Computing, DevOps, and Cybersecurity.
- **Hobbies:** Watching Cricket, Movies and TV Shows, Playing Chess, and Listening to Music.
- **Languages:** English, Hindi, Sindhi, Spanish.